

LOVIISA ORRENGRUND, Finland

WMO: 029920

Lat: 60.275N

Lon: 26.444E

Elev: 11

StdP: 14.69

Time Zone: 2.00 (EUE)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-0.7	4.2	-4.9	4.2	-0.5	0.6	5.7	4.8	35.8	28.1	33.3	28.6	12.9	10	0.784

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.6	75.5	69.0	73.0	67.1	70.3	64.8	69.9	74.4	68.0	72.6	65.5	69.5	10.4	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.1	103.4	73.4	66.1	96.1	71.6	63.5	87.7	68.6	33.9	74.6	32.4	72.7	30.3	69.7	74.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 34.2	30.8	27.7	DB	-1.2	77.0	9.1	4.4	-7.8	80.1	-13.1	82.7	-18.2	85.2	-24.8	88.3	(4)
(5)			WB	-1.7	69.8	9.0	3.2	-8.2	72.1	-13.5	74.0	-18.5	75.8	-25.0	78.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	43.4	25.6	23.7	28.4	37.7	48.6	56.5	64.1	62.9	55.6	45.6	38.1	32.7	
	DBStd	15.17	9.29	10.02	7.36	4.47	6.06	4.63	5.27	4.41	4.60	5.50	6.16	7.13	
	HDD50	3698	757	738	669	370	100	5	0	0	10	154	358	537	
	HDD65	7970	1222	1158	1134	819	508	258	83	95	282	601	808	1002	
	CDD50	1287	0	0	0	0	57	199	436	399	178	18	0	0	
	CDD65	85	0	0	0	0	0	2	54	29	0	0	0	0	
	CDH74	143	0	0	0	0	0	2	116	25	0	0	0	0	
	CDH80	3	0	0	0	0	0	0	3	0	0	0	0	0	
Wind	WSAvg	14.5	16.6	14.4	14.3	12.5	12.1	12.4	11.4	12.4	14.7	16.5	18.3	18.9	
Precipitation	PrecAvg	24.3	2.0	1.7	1.4	1.3	1.5	2.4	2.4	2.4	2.2	2.6	2.4	2.0	
	PrecMax	29.8	3.9	4.3	2.8	2.9	3.1	5.0	5.2	5.1	4.3	7.4	5.9	4.4	
	PrecMin	18.0	0.1	0.0	0.2	0.4	0.2	0.2	0.3	0.0	0.3	1.0	0.0	0.2	
	PrecStd	3.0	1.0	1.1	0.6	0.6	0.9	1.3	1.5	1.3	1.3	1.5	1.3	0.9	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.5	38.1	41.9	54.1	67.4	72.0	79.8	77.0	66.3	57.0	49.7	45.6	
		MCWB	39.2	35.5	37.8	46.8	57.8	63.7	69.8	70.4	62.5	54.7	48.3	44.2	
	2%	DB	39.0	36.0	38.8	49.3	63.7	68.5	77.1	73.8	64.3	55.6	48.4	43.8	
		MCWB	38.2	34.8	35.7	44.0	55.7	60.9	69.7	67.9	60.6	53.8	47.2	42.3	
	5%	DB	37.6	35.1	37.3	45.8	60.7	65.8	74.8	71.4	62.9	54.5	47.0	42.4	
		MCWB	36.6	34.1	35.1	41.3	54.4	59.4	68.6	65.6	59.3	52.7	45.8	41.1	
	10%	DB	36.2	34.1	35.9	43.6	57.6	63.4	72.4	69.2	61.7	53.0	45.5	40.7	
		MCWB	35.3	33.4	34.3	39.9	52.5	58.0	67.0	63.8	58.4	51.2	44.2	39.5	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.8	36.1	38.7	47.3	59.4	65.8	72.4	71.4	63.5	55.6	48.7	44.8
			MCDB	40.3	36.7	40.8	52.7	64.8	69.9	77.0	75.7	65.3	56.3	49.1	45.5
2%		WB	38.4	35.2	36.6	44.4	57.1	62.1	70.8	69.0	61.7	54.4	47.4	42.7	
		MCDB	38.8	35.8	37.9	48.8	62.1	66.8	75.3	73.0	63.5	55.3	48.1	43.5	
5%		WB	36.9	34.4	35.5	42.2	55.0	60.3	69.3	66.6	60.3	53.1	46.0	41.2	
		MCDB	37.4	35.0	36.7	45.0	59.5	64.8	73.9	70.3	62.2	54.2	46.8	42.1	
10%		WB	35.5	33.5	34.6	40.6	52.8	58.5	67.6	64.6	59.1	51.6	44.4	39.7	
		MCDB	36.1	34.1	35.6	42.8	57.3	62.8	72.4	67.9	61.3	52.9	45.2	40.6	
Mean Daily Temperature Range		5% DB	MDBR	6.6	6.6	7.3	7.0	8.3	8.1	7.6	6.7	5.6	5.1	4.9	5.5
			MCDBR	5.1	4.1	6.4	10.0	11.8	10.8	8.9	7.9	6.3	4.9	4.6	5.2
	MCWBR		5.3	3.9	4.7	5.9	6.8	6.2	5.2	5.0	4.7	4.9	5.1	5.3	
	5% WB	MCDBR	4.9	4.1	5.8	9.1	10.7	10.1	8.4	7.5	5.6	4.9	4.6	4.8	
		MCWBR	5.2	3.9	4.5	5.8	7.1	6.5	5.3	5.1	4.7	5.2	5.1	5.1	
Clear-Sky Solar Irradiance	taub	0.280	0.312	0.334	0.371	0.368	0.365	0.386	0.379	0.353	0.335	0.307	0.385		
	taud	2.138	2.168	2.212	2.284	2.402	2.460	2.420	2.435	2.491	2.385	2.210	2.601		
	Ebn at Noon	148	211	248	261	272	275	265	257	246	206	137	104		
	Edh at Noon	14	23	31	35	33	32	32	30	24	19	13	9		
All-Sky Solar Radiation	RadAvg	79	271	742	1226	1682	1750	1743	1332	794	329	89	43		
	RadStd	16	56	94	133	181	121	157	137	94	49	15	6		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (54 neighbors)	+0.85	N/A	N/A	N/A	N/A	N/A	-279	-297	N/A	N/A

Nomenclature: See separate page